

Sungmin Kim, Ph.D.



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<https://renariodyne.github.io/>

Research interests:

- Design of efficient algorithms and data structures
- Automata and formal languages

Education

2022.03 – 2026.08



Ph.D. in Computer Science, Yonsei University

Ph.D. supervisor: Prof. Yo-Sub Han

Dissertation title: Pattern matching under Simon's congruence

2017.03 – 2022.02



B.A. in English Language and Literature, Yonsei University



B.S.E. in Computer Science, Yonsei University

Double major, Highest honors at graduation. GPA: 4.26 / 4.3

Research Publications

Journal Articles

- 1 P. Fleischmann et al., "Matching patterns with variables under Simon's congruence," *Journal of Automata, Languages and Combinatorics*, 2026, to appear.
- 2 **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," *Information and Computation*, 2026, to appear.
- 3 Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, "Existential and universal width of alternating finite automata," *Information and Computation*, p. 105 337, 2025.
- 4 **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," *Theoretical Computer Science*, p. 115 506, 2025.
- 5 **S. Kim**, S.-K. Ko, and Y.-S. Han, "Simon's congruence pattern matching," *Theoretical Computer Science*, vol. 994, p. 114 478, 2024.
- 6 **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On Simon's congruence closure of a string," *Theoretical Computer Science*, vol. 972, p. 114 078, 2023.

Conference Proceedings

- 1 **S. Kim**, H. Hong, T. Lim, Y.-S. Han, and K. Salomaa, "Decomposing regular languages under shuffle along trajectories," in *Proceedings of the 30th International Conference on Implementation and Application of Automata (CIAA)*, to appear, 2026.
- 2 **S. Kim**, H. Jin, and Y.-S. Han, "Pattern matching under $\mathcal{R}$ -congruence," in *Proceedings of the 30th International Conference on Implementation and Application of Automata (CIAA)*, to appear, 2026.
- 3 **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," in *Proceedings of the 29th International Conference on Developments in Language Theory (DLT)*, vol. 16036, 2025, pp. 182–196.
- 4 **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," in *Proceedings of the 28th International Conference on Developments in Language Theory (DLT)*, vol. 14791, 2024, pp. 189–202.
- 5 P. Fleischmann et al., "Matching patterns with variables under Simon's congruence," in *Proceedings of the 17th International Conference on Reachability Problems (RP)*, vol. 14235, 2023, pp. 155–170.

- 6 Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, “Existential and universal width of alternating finite automata,” in *Proceedings of the 25th International Conference on Descriptive Complexity of Formal Systems (DCFS)*, vol. 13918, 2023, pp. 51–64.
- 7 **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, “On the Simon’s congruence neighborhood of languages,” in *Proceedings of the 27th International Conference on Developments in Language Theory (DLT)*, vol. 13911, 2023, pp. 168–181.
- 8 **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, “On Simon’s congruence closure of a string,” in *Proceedings of the 24th International Conference on Descriptive Complexity of Formal Systems (DCFS)*, vol. 13439, 2022, pp. 127–141.
- 9 **S. Kim**, S.-K. Ko, and Y.-S. Han, “Simon’s congruence pattern matching,” in *Proceedings of the 33rd International Symposium on Algorithms and Computation (ISAAC)*, vol. 248, 2022, 60:1–60:17.

Under Review

- 1 **S. Kim** and Y.-S. Han, “Approximate pattern matching under Simon’s congruence,” submitted to a conference.
- 2 **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, “On the Simon’s congruence neighborhood of languages,” submitted to Theoretical Computer Science.

Talks and Seminars

DLT’25	■	Pattern mining under Simon’s congruence.
DLT’24	■	Approximate Cartesian tree pattern matching.
WAAC’24	■	Approximate Cartesian tree pattern matching.
DCFS’23	■	Existential and universal width of alternating finite automata.
DLT’23	■	On the Simon’s congruence neighborhood of languages.
FWAC’23	■	On the Simon’s congruence neighborhood of languages.
ISAAC’22	■	Simon’s congruence pattern matching.
DCFS’22	■	On Simon’s congruence closure of a string.

Academic Activities and Services

Reviewer	■	Conferences: DLT’26 / CIAA’26 / DCFS’26 / STACS’26 / CIAA’25 / DLT’25 / DCFS’25 / ISAAC’24 / DLT’24 / CIAA’24
	■	Journals: Computer Science Review / Journal of Automata, Languages and Combinatorics / Theoretical Computer Science / International Journal of Foundations of Computer Science
Teaching Assistant	■	Data Structures / Automata and Formal Languages / Theory of Computation / Analysis of Algorithms

Experience

2026.10 –	■	Postdoctoral researcher at Göttingen University, Germany
2025 – 2026	■	Participation in Industry–University Collaborative Project, LG
Collaboration	■	Göttingen University, Germany / Queen’s University, Canada

Miscellaneous

Other Skills

Languages	📖 Korean (native speaker), English (fluent)
Programming	📖 C++, Java, Python

Scholarships

2026	📖 Postdoc Fellowship at the University of Göttingen
2022 – 2024	📖 Accelerated degree program scholarship
2023	📖 Academic research support ARF fellowship grant
2017–2021	📖 Performance and merit scholarships during B.A./B.S.E. studies

As of: July, 2026